

Experiment: To find index of Prism by critical angle method

Paratus:

Glass Prism

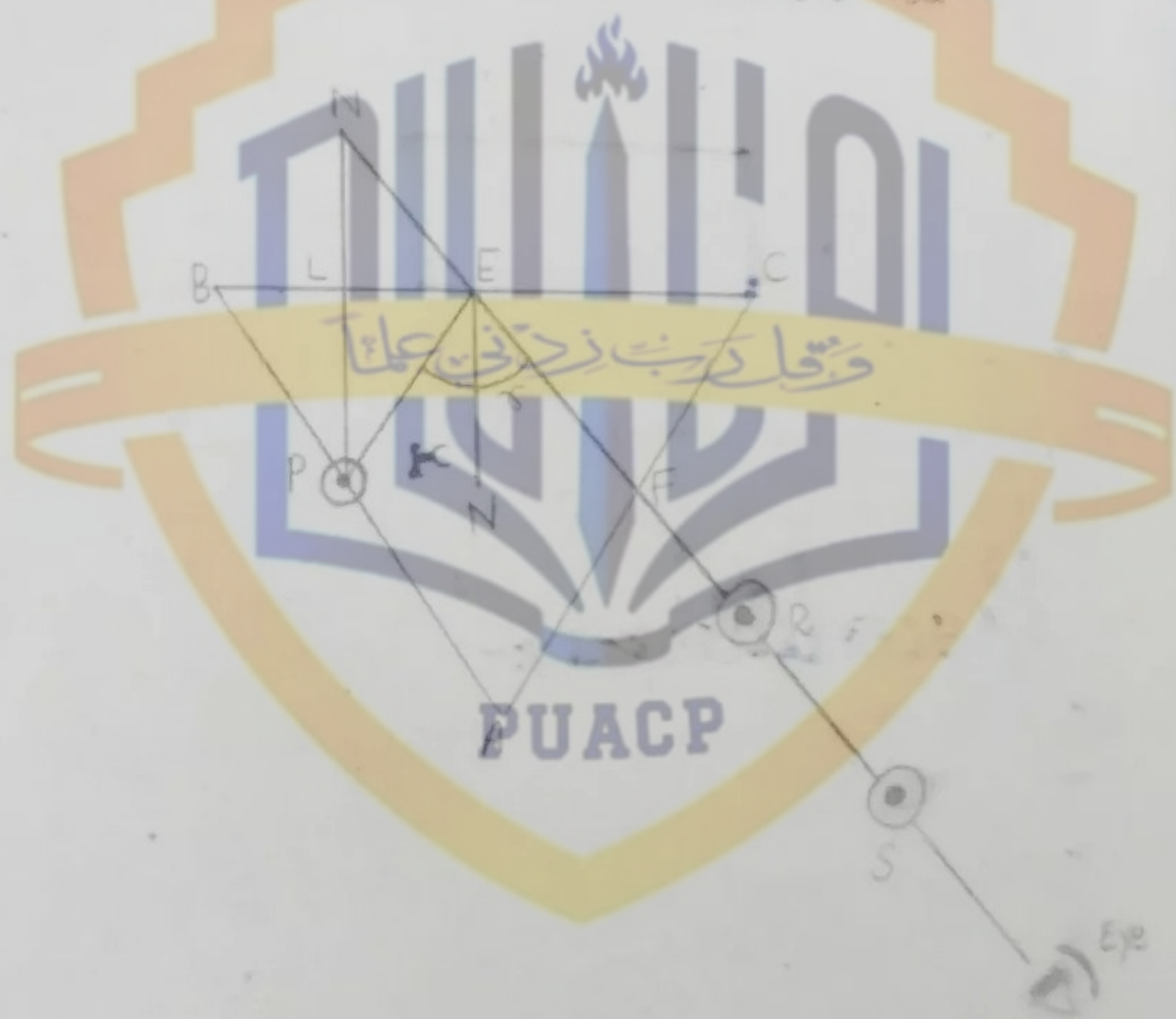
White sheet

• Wooden board

• Pencil

• Protrector

• meter rod



Experiment Name: **To find index of Prism by critical angle method**

Apparatus:-

- Glass Prism
- White sheet
- Wooden board
- Pencil
- Protractor
- meter rod

Procedure:-

- Fix white sheet of paper on drawing board with fixing pins.
- Remove prism and fix a pin P just over the boundary line AB in the centre.
- replace the Prism in its previous position, so that Pin P just touches the face AB.
- keeping the image in view, move your eye from C towards A, the image will become faint and will disappear.
- Remove the Prism and pin after making their positions.
- Join pin pitch R and S to meet face AC in F.
- Draw PL perpendicular to BC produce PL to N such that $PL = LN$. Join N to F cutting BC at E. Join PE and measure angle $\angle PEF$. Also angle $\angle$

Calculation and Observation:

no. of Obs.	$\angle$ PEF	Critical angle $\angle C = \frac{1}{2} (\angle \text{PEF})$
1.	80°	$\frac{1}{2} (80^\circ) = 40^\circ$
2.	84°	$\frac{1}{2} (84^\circ) = 42^\circ$
3.	85°	$\frac{1}{2} (85^\circ) = 42.5^\circ$

Refracting index of glass: $\mu = \frac{1}{\sin C}$

$$= \frac{1}{\sin 42^\circ}$$

$$= \frac{1}{0.6691}$$

$$= 1.494 = 1.5$$

Actual Value of $\mu = 1.50$

Result:

Refractive index of glass: 1.5

$$\angle PEN' = \angle FEN' = \text{critical angle}$$

- Repeat the experience thrice and find mean value of critical angle C .

Precautions:-

- The faces of Prism should be clean.
- Pin should be fix erect.
- The base of the prism be away from the observer.
- The edge A of prism should be fixed in line with the point of pin P or when it is just on the point of disappearance.
- The path of light should be shown by arrow head.